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Research Article

Temporal asynchrony affects letter-speech sound integration in second language reading

Dongyang Yan [a,*], Ayumi Seki [b]

[a] Graduate School of Education, Hokkaido University, Sapporo, Japan
[b] Faculty of Education, Hokkaido University, Sapporo, Japan

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ABSTRACT

The integration of visual letters with corresponding auditory speech sounds is crucial for literacy acquisition in alphabetic languages. Previous research exploring this process has shown that audiovisual (AV) congruency enhances auditory N1 amplitude, particularly in advanced English learners, during second language (L2) reading. In the current study, we investigated how the degree of temporal coincidence between letters and speech sounds influences AV congruency effects, using both behavioral and event-related measures. Native Japanese speakers with an intermediate level of English proficiency completed a priming task in English. During the task, visual letters or symbols preceded auditory sounds across different stimulus onset asynchrony (SOA) conditions. The behavioral experiment revealed shorter response times for congruent trials across all SOA conditions, indicating a facilitation effect of congruency. In the electroencephalography experiment, we found an increased early N1 response to congruent AV pairs in the bilateral frontotemporal scalp region during L2 reading, but only in the AV300 condition (with letters preceding speech sounds by 300 ms). Our findings highlight the importance of temporal relationship on N1 congruency effects. Specifically, a slightly longer SOA may help compensate for lower levels of automaticity in intermediate L2 learners, thus facilitating successful integration of letters and sounds during English reading. This study contributes not only to a better understanding of L2 reading mechanisms, but also provides a theoretical basis for examining the role of atypical letter-speech sound integration related to L2 reading difficulties, particularly those with limited proficiency.

1. Introduction

Humans live in a complex environment rich in signals from multiple sensory modalities, requiring the brain to integrate these multisensory inputs effectively to construct coherent perceptual representations. Among the various forms of multisensory integration, the integration of visual letters with corresponding auditory speech sounds is considered as a fundamental mechanism in literacy acquisition (Ehri, 2005). Through repeated exposure and practice, learners gradually form letter-speech sound (L-SS) associations. For skilled readers, these associations become highly overlearned and function similarly to other natural multisensory representations. However, it takes much longer for these associations to support automatic integration of letters and speech sounds. Fluent reading, in particular, depends heavily on the automatic activation of established L-SS associations, and failure in this process is often linked to reading difficulties (Blomert, 2011). As such, this L-SS integration may be rather influential in successful and impaired reading development in native and second language learners.

A basic understanding of the neural network for L-SS integration was reported by manipulating the congruency of letter-sound pairs. In this paradigm, researchers typically compare the responses to audiovisual (AV) stimuli that are either previously associated (e.g., the visual letter “d” and its congruent speech sound /d/) or unassociated (e.g., the visual letter “d” and an incongruent pronunciation /k/). Across studies, significantly stronger neural responses have consistently been reported for congruent than for incongruent AV pairs. The greater neural responses to congruent pairs—commonly referred to as the “congruency effect”—are interpreted as reflecting successful integration of visual and auditory linguistic information. Using this congruency effect, previous neuroimaging studies (e.g., Blau et al., 2010; Blau, Van Atteveldt, Ekkebus, Goebel, & Blomert, 2009; Blau, Van Atteveldt, Formisano, Goebel, & Blomert, 2008; Raij, Uutela, & Hari, 2000; van Atteveldt, Formisano, Goebel, & Blomert, 2004; van Wassenhove, Grant, & Poeppel, 2007) have revealed the involvement of heteromodal regions

* Corresponding author at: Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan.
E-mail address: ydyxj@hotmail.com (D. Yan).

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(i.e., the superior temporal cortex [STC]) as well as modality-specific areas, including the auditory cortex (Heschl's sulcus [HS]; planum temporale [PT]), in L-SS integration. A review by van Atteveldt, Roebroeck, and Goebel (2009) proposed the feedback hypothesis, which further suggests that the integration of letters and speech sounds recruits a broader network, and that the outcomes of this integration would be primarily fed back to low-level auditory processing and modulate activities within the auditory cortex.

Importantly, the efficiency of AV integration is strongly influenced the temporal relationship between inputs from AV modalities (Adams, 2016; Calvert, Spence, & Stein, 2004; van Wassenhove et al., 2007). In natural environments, the integration of multisensory signals into a unified percept often depends on the temporal proximity between modalities, rather than on their exact physical simultaneity. This is because differences in physical transmission speed and neural processing characteristics across sensory pathways can cause auditory and visual inputs to be processed at different rates. As a result, optimal multisensory integration may occur at a specific temporal interval between modalities, also referred to as the point of subjective simultaneity (Keetels & Vroomen, 2012). Within this context, there exists a limited time window for effective multisensory integration, known as the temporal binding window (TBW) (Colonius & Diederich, 2004). TBW defines the range of temporal relationships across modalities within which sensory inputs are likely to be integrated into a coherent percept. Accordingly, both the position of the point of subjective simultaneity and the width of TBW are considered to be important factors when assessing changes in the TBW.

Similar to other forms of multisensory integration, learning to read is a temporally constrained process that requires rapid and precise matching of visual letters with correct speech sounds within the limited binding window. Individuals with reading difficulties exhibit an abnormally enlarged TBW during general AV integration (e.g., Hairston, Burdette, Flowers, Wood, & Wallace, 2005; Laasonen, Service, & Virsu, 2002; Virsu, Lahti-Nuuttila, & Laasonen, 2003). Moreover, previous studies indicate that the size of the TBW can vary depending on prior learning experiences (Powers 3rd, Hillock, & Wallace, 2009; Theves, Chan, Naumer, & Kaiser, 2020). Based on these findings, it is reasonable to assume that differences in language experience and proficiency levels may influence TBW in L-SS integration.

Studies supporting this assumption have explored how the size of TBW influences L-SS integration in proficient native language (L1) learners, typically by manipulating the stimulus onset asynchrony (SOA) between letters and speech sounds. Using the congruency effect as an indicator, an earlier neuroimaging study by van Atteveldt, Formisano, Blomert, and Goebel (2007) explore the impacts of temporal asynchrony between letter-sound pairs on L-SS integration using different SOAs (i.e., speech sounds appear ±300 ms before visual letters). The results indicated that the congruency effects in the auditory cortex and STC fluctuated depending on different SOAs. Notably, the HS/PT only showed stronger congruency effects when auditory and visual stimuli were presented synchronously. These findings highlight the influence of temporal relationship between AV inputs on L-SS integration during alphabetic L1 reading.

Given its high temporal resolution, event-related potential (ERP) has been also used to investigate the precise time course of L-SS integration. A prominent approach is the mismatch negativity (MMN) paradigm, in which a rare deviant bimodal stimulus (e.g., visual letter “a” paired with auditory /o/, incongruent) is embedded within a sequence of frequent standard AV stimuli (e.g., visual “a” paired with auditory /a/, congruent). In this context, bimodal MMN reflects an automatic neural response to violations of learned AV associations, often elicited by a bimodal deviant stimulus. Using this paradigm, a series of studies have manipulated SOAs to investigate their impact on L-SS integration in typical adult readers (Froyen, Van Atteveldt, Bonte, & Blomert, 2008), typically developing children (Froyen, Bonte, van Atteveldt, & Blomert, 2009) and children with dyslexia (Froyen, Willems, & Blomert, 2011; Žarić et al., 2014). Froyen et al. (2008) compared auditory-only MMN responses with those elicited by simultaneously presented AV stimuli (AV0) and by bimodal AV stimuli with delayed speech sounds (AV200) in typical adults. Their findings revealed an enhanced bimodal MMN to the cross-modal deviant sound in the AV0 condition, emerging approximately 100–250 ms after auditory onset, whereas no such enhancement was found in the AV200 condition. Although derived from a different rationale than the congruency effect, the enhancement of bimodal MMN is similarly interpreted as an index of automatic L-SS integration. Meanwhile, typically developing children demonstrated an enhanced bimodal MMN only in the AV200 condition (Froyen et al., 2009). Younger children did not show the early enhancement in either the AV0 or AV200 condition, but demonstrated an increased late bimodal MMN at approximately 600 ms only in the AV200 condition (Žarić et al., 2014). These findings suggest that TBW for L-SS integration is larger in typically developing children than in adults. Moreover, children with dyslexia may require an even wider TBW than their typically developing peers.

Building on these MMN findings, a more recent ERP study by Du, Li, Qin, and Bi (2022) used congruency effects as an indicator of automatic L-SS integration. Their study focused on non-alphabetic Chinese children with and without dyslexia and revealed an auditory N1/P2 congruency effect in the AV0 condition for typically developing readers. In contrast, the dyslexic group exhibited a congruency effect only in the AV300 condition (letter preceding the speech sound by 300 ms). Collectively, previous findings on L1 reading indicate the critical role of temporal relationships in the integration of letters and speech sounds not only for alphabetic, but also for non-alphabetic L1 reading. Additionally, these studies on L1 reading revealed the influence of reading ability on the temporal window for integration, suggesting that individuals with lower language proficiency may require a larger TBW for integration (Du et al., 2022; Froyen et al., 2008; Froyen et al., 2009; Froyen et al., 2011; Žarić et al., 2014).

Importantly, however, the developmental pathways through which L-SS associations may be established differ between L1 and L2 learners. Unlike L1 learners, who typically develop relatively stable phonological knowledge before the acquisition of grapheme-phoneme correspondences (Phillips, Menchetti, & Lonigan, 2008), L2 learners often encounter phonemic sounds and orthographic input simultaneously while they are still establishing new phonetic categories. In this context, orthographic forms (letters) may play a particularly important role during unfamiliar L2 phonological learning, as L2 learners often rely more heavily on visual letter cues to support the acquisition of novel speech sounds (Hayes-Harb & Barrios, 2021). This greater reliance on orthography may shape the formation of L-SS associations in ways that differ from L1 learning, and may consequently influence the temporal dynamics of L-SS integration. This distinction may provide an important motivation for examining the role of temporal relationships in L-SS integration in L2 learners, rather than directly extrapolating findings from L1 populations.

Consistent with this perspective, recent research has focused on exploring how temporal relationship between modalities impacts L-SS integration in second language (L2) reading. Wang, Yang, Cao, Liu, & Tao (2019), Wang et al. (2022) and Du, Yang, Wang, Li, and Tao (2023) investigated this integration process in L2 among native Chinese speakers with varying English proficiency levels. Their findings consistently revealed a lack of bimodal MMN enhancement in native Chinese speakers with intermediate English levels in either the AV0 or AV200 condition. However, native Chinese speakers with more extensive exposure to English exhibited a bimodal MMN enhancement in the AV0 condition, similar to that observed in native English peers. Their findings may highlight the importance of temporal relationship between AV pairs in L2 L-SS integration among individuals with different reading experiences.

Complementing this line of work, our recent study (Yan & Seki, 2024) employed a priming paradigm to investigate L2 L-SS integration processing in native Japanese learners of English. The stronger congruency effects were observed within N1 time window for advanced Japanese learners of English, when visual stimuli preceded auditory stimuli by 200 ms. In contrast, intermediate L2 learners did not exhibit significant congruency effects. Importantly, the greater neural response to congruent AV pairs compared with incongruent pairs was significantly correlated with the English proficiency levels, suggesting that early congruency effects may reflect the automatic activation of L-SS associations, which emerges with increased language proficiency. Considering the role of the temporal relationship between auditory and visual inputs in successful L-SS integration, the absence of congruency effects in intermediate learners may result from an insufficient time window for the integration of English letters and sounds. Taken together, evidence from both L1 and L2 reading studies suggests that the temporal window required for successful L-SS integration narrows as language experience increases. Based on this, we hypothesized that intermediate English (IE) learners may potentially require a slightly longer interval for the integration of English letters and sounds, than advanced learners. However, this hypothesis, particularly regarding the timing constraints of L-SS integration in intermediate L2 readers, remains unexamined and warrants further investigation.

Building upon prior research that primarily focused on the width of the TBW in reading, the aim of the current study was to explore how different SOAs affect the integration of English letters and sounds in native Japanese speakers with an intermediate level of English proficiency. Specifically, we focused on early congruency effects within the auditory N1 time window and investigated their spatial scalp distribution. Three types of SOAs were included as the asynchronous condition: AV200 (visual letters preceding auditory sounds by 200 ms), AV300, and AV600. The choice of AV200 condition aimed to replicate our previous study (Yan & Seki, 2024), in which no reliable N1 congruency effect was observed in IE learners. Additionally, based on previous studies (Du et al., 2022; Peter, Porada, Regenbogen, Olsson, & Lundström, 2019; Theves et al., 2020; van Atteveldt et al., 2007), the proportion of perceived simultaneity between auditory and visual stimuli dramatically decreased at an SOA of 300 ms among healthy adults. Therefore, we selected the AV300 condition as one type of SOA condition, as we expected that a slightly extended TBW would facilitate English L-SS integration in IE learners, as evidenced by greater neural responses to congruent AV stimuli. The AV600 condition was selected to assess whether a further increase in SOA would lead to the perception of auditory and visual inputs as separate events, thereby reducing the likelihood of integration (Leone & McCourt, 2015). In line with our previous study, we hypothesized that congruency effects in early ERP responses would be found only when visual stimuli preceded auditory stimuli by 300 ms, but not by 200 ms. Furthermore, given that a further extended SOA may increase the likelihood of perceiving auditory and visual stimuli as separate events, we predicted the absence of congruency effects in the early auditory N1 response for the AV600 condition. Finally, motivated by our previous findings that early N1 congruency effects were associated with English proficiency, we supplementarily explore the potential relationships between English proficiency levels and audiovisual congruency effects within each SOA condition.

2. Materials & methods

2.1. Participants

Twenty-five native Japanese-speaking university students aged 18–22 years participated in this study. The sample size was determined based on data from a previous study (Yan & Seki, 2024), which employed a similar paradigm and demonstrated reliable effects with a comparable sample size. Participants were recruited according to their TOEFL ITP scores obtained within two years, which ranged from 400 to 600, generally indicating intermediate levels of English proficiency, also with English proficiency levels equivalent to those of IE group in our previous study. All participants started learning English after the entry to primary school and had no prior experience living in English-speaking countries. Additionally, they used English as their L2 and had no exposure to a third language. Three participants did not attend the electroencephalogram (EEG) experiment and were therefore excluded from the analysis, resulting in a final sample of 22 participants who completed both the behavioral and EEG experiments. Of these, all 22 participants were included in the behavioral analyses, whereas two participants were excluded from the ERP analyses due to more than 30% invalid trials, yielding a final ERP sample of 20 participants. The study procedure was approved by the Ethics Committee of the Department of Education at Hokkaido University and was conducted in accordance with the Declaration of Helsinki. All the participants provided written informed consent. All the participants were right-handed, had normal or corrected-to-normal vision, and reported no hearing impairments, reading difficulties, or other neurological or psychiatric disorders.

2.2. Stimuli and procedure

Before attending the laboratory for the study, all participants were asked to completed an online form detailing their language background. The behavioral and EEG experiments were conducted on two separate days. Before the commencement of the experiments, the participants received a tour of the laboratory facilities and a brief introduction on the experimental procedures. On the initial first day, they completed a 60-min pre-test, including a batch of standardized language proficiency measures in both Japanese and English, as well as assessments of general cognitive ability. Subsequently, a 15-min behavioral letter-sound priming experiment was conducted. The EEG priming experiment was performed on a separate day, typically scheduled within 7 days of the initial behavioral assessment (range: 2–7 days; mean = 4 days, standard deviation [SD] = ±1.92), and lasted approximately 50 min.

2.2.1. Measurement of L1 & L2 proficiency and cognitive ability

We used three standardized English literacy tests to measure participants' basic reading abilities in English. Demographic information, including age, sex distribution, and performance on behavioral pre-tests, is summarized in Table 1.

Table 1
Demographic information and mean (SD) of scores on behavioral English proficiency and general cognitive ability for IE learners.

Measure (Points)	Intermediate English (IE) Learners
	Mean (SD)
N (male/female)	22(12/10)
Age (years)	19.64(1.18)
Age of L2 acquisition (years)	11.41(2.15)
TOWRE II Sight Word Efficiency raw score	73.73(8.11)
TOWRE II Phonemic Decoding Efficiency raw score	38.09(4.89)
WJ III LWI raw score	51.59(2.15)
WJ III WA raw score	25.09(2.60)
WJ III RF raw score	45.45(8.20)
WJ III SS raw score	19.50(2.70)
CTOPP-2 Blending Words raw score	18.82(5.77)
CTOPP-2 Blending Nonwords raw score	13.00(3.89)
CTOPP-2 Segmenting Nonwords raw score	18.09(4.68)
RaWF test Japanese letter counts	723.50(135.48)
ASRS checklist rating raw score	26.95(13.73)
Matrix reasoning test raw score	21.00(2.05)

For each English ability subtest, the total number of correctly answered questions was used as the raw score, except for the WJ III SS test.
TOWRE = Test of Word Reading Efficiency; WJ = Woodcock–Johnson; LWI = Letter-Word Identification; WA = Word Attack; RF = Reading Fluency; SS = Spelling of Sounds; CTOPP = Comprehensive Test of Phonological Processing; RaWF = Reading and Writing Fluency Task for Japanese University students; ASRS = Adult ADHD Self-Report Scale.

2.2.1.1. Test of Word Reading Efficiency II. Participants were instructed to complete the Sight Word Efficiency and Phonemic Decoding Efficiency subtests from the Test of Word Reading Efficiency II (Torgeson, Wagner, & Rashotte, 1999). These subtests required to read aloud a list of 108 words and 66 pronounceable nonwords as quickly and accurately as possible within a time limit of 45 s.

2.2.1.2. Woodcock–Johnson III (WJ III). Participants completed the Letter-Word Identification (76 words) and Word Attack (32 pseudowords) subtests from the WJ III (Schrank, 2011) by providing the correct pronunciation of these words and pseudowords with no time limit. Each test was discontinued after six consecutive errors or no responses.

The Reading Fluency test was used to assess English reading fluency. Participants evaluated a series of short sentences and marked "Yes" if the sentence was true, or "No" if it was a paradox. They were given a total of 3 min to complete as many items as possible, with a maximum of 98 questions.

Spelling ability was assessed using the Spelling of Sounds test. Participants listened to an audio recording of pronounceable nonwords twice and wrote down the correct spellings. The test contains 28 nonwords and ended after four consecutive errors or no responses. The maximum possible score was 45 points.

2.2.1.3. Comprehensive Test of Phonological Processing 2nd Edition (CTOPP-2). The CTOPP-2 (Wagner, Torgesen, Rashotte, & Pearson, 2013) was used to evaluate reading-related phonological processing skills. In the Blending Words and Blending Nonwords subtests, the participants synthesized sounds to form words or nonwords. In the Segmenting Nonwords subtest, the participants segmented phonemes from pronounceable nonwords. Each subtest ended after three consecutive errors or no responses.

2.2.1.4. Reading and Writing Fluency Task for Japanese university students (RaWF). The reading fluency task from the RaWF test (Takahashi & Mitani, 2022), with a 1-min time limit, was used to measure Japanese reading ability. Participants silently read each short sentence and judged it as either true or paradoxical by marking "×" or "〇". The task included a maximum of 55 items. We used the total Japanese letter counts (e.g., "か" /ka/ counted as a single letter) as a measure to ensure that none of the participants had reading difficulties in their native language. Reading speed of less than one SD below the mean (−1.0 SD; 357–418 Japanese letters or fewer) was considered indicative of slow silent reading in Japanese.

2.2.1.5. Matrix reasoning test. A nonverbal intelligence test from the Wechsler Abbreviated Scale of Intelligence (WASI; Wechsler, 1999) was used to assess general cognitive abilities. The matrix reasoning subtest assessed participants' ability to solve novel problems, identify relationships, and complete visual analogies, without assessing their vocabulary or language proficiency. This test comprised 26 questions and was discontinued after four consecutive errors or no responses. The total number of correctly answered questions was used as the raw score.

2.2.1.6. Attention Deficit Hyperactivity Disorder (ADHD) check list. Given that individuals with more ADHD-like traits may show an abnormal TBW during multisensory integration (Panagiotidi et al., 2017), we utilized the World Health Organization Adult ADHD Self-Report Scale (ASRS; Kessler et al., 2005) to assess ADHD-like traits among our participants. The ASRS contains 18 items from the DSM-IV-TR (American Psychiatric Association, 2013) to measure the frequency of ADHD symptoms over the past six months. Participants rated their experience using a five-point Likert scale, ranging from 0 ("never") to 4 ("very often") (Kessler et al., 2005). The ASRS is divided into two subscales: inattention and hyperactivity/impulsivity subscale, each comprising nine items (Reuter, Kirsch, & Hennig, 2006).

2.2.2. Behavioral priming measure

We applied an AV priming paradigm similar to that used in a previous study (Yan & Seki, 2024). To create experimental temporal asynchronies, we manipulated the AV stimulus by delaying the auditory stimulus. We created three types of asynchronies, with the visual letters leading the auditory sounds by 200 ms (AV200), 300 ms (AV300), and 600 ms (AV600). Participants completed a behavioral priming task under three conditions (i.e., congruent, incongruent, and control conditions) for each SOA type. In the congruent and incongruent conditions, the visual primes and auditory stimuli corresponded to matching and mismatching letter-speech sound pairs, respectively. In the control condition, participants were presented with a visual prime, while the auditory stimuli were composed of scrambled phonemes generated by randomly assembling 5 ms segments of the original speech phoneme using MATLAB code from Ellis (2010). The control condition was included to ensure balanced button press responses and to avoid the dominance of "yes" responses to speech sounds, although data from the control condition were excluded in the later analysis. Both the English visual and auditory stimuli were aligned with those used in a previous study (Yan & Seki, 2024). Durations for each sound are shown as follows: /d/ (171 ms), /k/ (143 ms), /p/ (147 ms), and /z/ (215 ms). Only consonant phonemes were used to minimize pronunciation variability that could arise from vowel sounds. All speech sounds were off-line down-sampled to 22.05 kHz and matched for loudness at 70 dB using the PRAAT software (www.fon.hum.uva.nl/praat/) (Boersma & Weenink, 2020). Visual stimuli were displayed on a 24-in. computer monitor (1920 × 1080 resolution) at a viewing distance of 60 cm. Auditory stimuli were presented to both ears via Panasonic headphones. Stimulus presentation and response collection were controlled using *E*-Prime software (version 3.0).

Participants were instructed to prompt to respond after hearing the auditory stimuli by indicating whether the sound was a "real speech sound", as shown in Fig. 1A. To familiarize themselves with the task, participants completed a performance-based practice session. Practice trials continued until an accuracy criterion of 80% was reached, resulting in a variable number of practice trials across participants; however, all participants needed to complete a minimum of 20 practice trials. In each trial, a central fixation was first displayed for a variable duration ranging from 1000 to 1450 ms in 150 ms increments to minimize expectancy effects. Subsequently, a black Arial visual letter (font size: 28) appeared on a gray screen for 200, 300, or 600 ms, followed by auditory stimuli. Each SOA condition comprised two blocks, and the participants completed each type in a pseudorandom sequence. Within one block, three conditions (i.e., congruent, incongruent, and control) were presented in random order, with 20 trials for each condition. Each block lasted approximately 2 min, and the entire experiment took approximately 15 min to complete. Participants were given an opportunity to rest after completing each block.

2.2.3. EEG priming measure

To mitigate the impact of task-related top-down processing on congruency effects, a passive task was employed during the EEG experiment (Fig. 1B). Participants were instructed to withhold physical responses to the AV stimuli and instead focus on detecting the presence of piano pictures (i.e., the target condition) to maintain attention. Visual stimuli were displayed at the center of a 24-in. widescreen monitor (Dell G2422HS) with a resolution of 1920 × 1080 pixels and a refresh rate of 165 Hz. During the EEG task, the participants sat in a sound-attenuating, electrically shielded room, positioned approximately 50 cm from the screen. Each block consisted of 64 trials including four conditions (target, congruent, incongruent, and control), which were presented with equal frequency and in a randomized order within the block. SOA was manipulated between blocks. To enhance the reliability of the ERP data, the EEG experiment included six blocks for each SOA condition, for a total of 96 trials for each condition. The order of SOA blocks was pseudorandomized and counterbalanced across participants. The entire

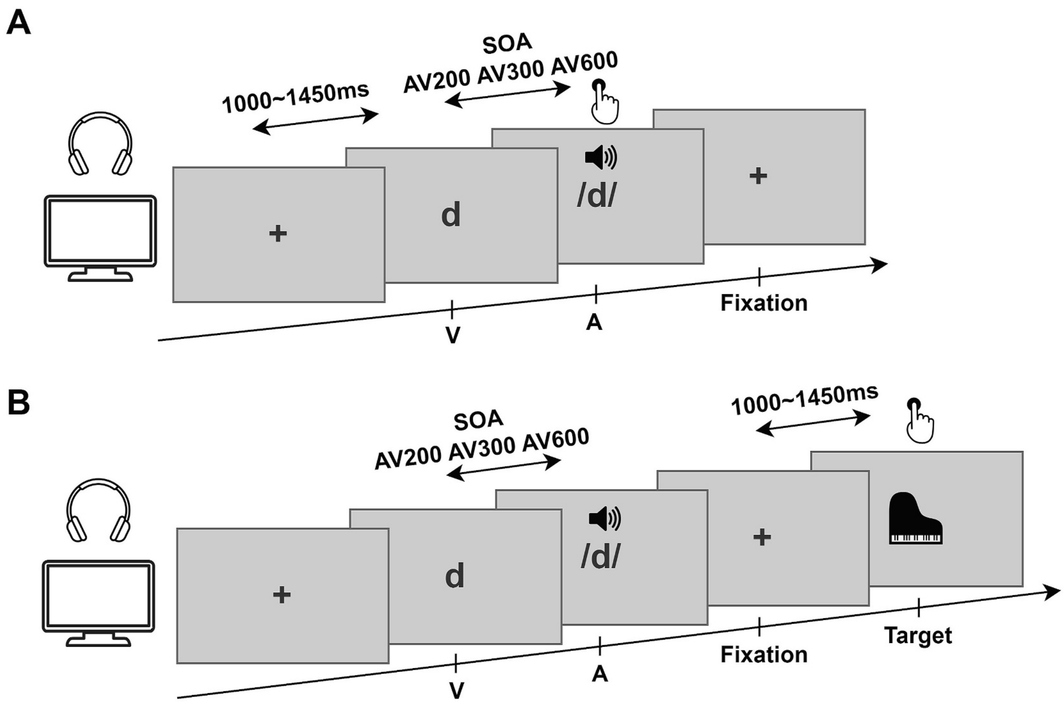

Fig. 1. Task flow for behavioral (**A**) and EEG (**B**) priming paradigms. SOA: stimulus onset asynchrony.

experiment lasted approximately 50 min, with breaks provided upon participants' request.

2.3. EEG recording and preprocessing

Data were recorded using a 29-channel active EEG system (EasyCap, Herrsching, Germany) as specified by the International 10–20 System (Fp1/2, F3/4, F7/8, Fz, FC1/2, FC5/6, T7/8, C3/4, Cz, CP5/6, CP1/2, P3/4, P7/8, Pz, POz, O1/2, and Oz). Three electrooculogram electrodes were simultaneously used to monitor eye movements. POz served as the reference electrode. All data were amplified using a SynAmps amplifier (NeuroScan, Sterling, VA, USA) and digitized at a sampling rate of 1000 Hz. Electrode impedance was maintained below 10 kΩ.

Raw data were analyzed using EEGLAB 14.1.2b (Delorme & Makeig, 2004), ERPLAB Toolbox (Lopez-Calderon & Luck, 2014), and custom-made MATLAB scripts (MathWorks, Natick, MA, USA). All EEG signals were down-sampled to 250 Hz. Continuous EEG data were filtered with a high-pass filter (infinite impulse response filter, half-amplitude cutoff = 0.1 Hz, 12 dB/oct roll-off). Epochs ranging from 900 ms before to 800 ms after the onset of auditory sounds were then extracted for data analysis. Independent components associated with artifacts were identified using ICLabel (Pion-Tonachini, Kreutz-Delgado, & Makeig, 2019) and subsequently removed. Channels containing large artifacts were excluded and reconstructed through spherical interpolation (on average 1.40 ± 1.21 channels per participant across all runs). The epoched data were then re-referenced to a common average. Trials containing remaining artifacts of amplitudes ± 100 μV at scalp electrodes were excluded. Data from two participants were excluded from further EEG analyses due to more than 30% of the trials were rejected. The average number (SD) of epochs in each condition per SOA condition is shown in Table 2. Before the calculation of ERP, all epoched data were low-pass filtered at 35 Hz using an infinite impulse response filter and baseline-corrected using the −100 to 0 ms interval before sound onset for each SOA condition.

Table 2
Average number (SD) of epochs per SOA condition in each condition.

	AV200	AV300	AV600
AVc	83.65 (±9.28)	83.6 (±8.80)	85.2 (±7.92)
AVi	83.85 (±8.70)	83.9 (±8.55)	85.4 (±8.29)

AVc = audiovisual congruent; AVi = audiovisual incongruent.

2.4. Statistical analysis

2.4.1. Data from behavioral experiment

We analyzed accuracy (percentages) and RTs (milliseconds) for each participant in the behavioral priming task. A 2 × 3 repeated-measures analysis of variance (ANOVA) was conducted to evaluate the influence of conditions (congruent, incongruent) and SOA conditions (AV200, AV300, and AV600) on RTs. Normality was examined using QQ plots. Statistical significance was set at a threshold of $p < 0.05$. When the assumption of sphericity was violated, Greenhouse–Geisser corrections were applied to adjust the degrees of freedom in the repeated-measures ANOVA. The jamovi project (2024) was used for all statistical analyses.

2.4.2. Data from EEG experiment

To ensure participants' engagement and compliance during the EEG priming task, we calculated the average accuracy and RTs for target detection. Statistical analyses were then conducted on epochs time-locked to the auditory onset within the N1/P2 time windows. As previous studies on typical topographical distribution have suggested AV congruency effects mostly observed at central sites (Knowland, Mercure, Karmiloff-Smith, Dick, & Thomas, 2014; Stekelenburg & Vroomen, 2007); thus, N1/P2 peak latencies and mean amplitudes at electrode Cz were measured for each SOA condition to assess the influence of different SOAs on congruency effects in intermediate L2 English learners. To obtain more accurate and robust estimates of component latencies, we employed a jackknife-based technique (Kiesel, Miller, Jolicoeur, & Brisson, 2008; Miller, Ulrich, & Schwarz, 2009). Specifically, peak latencies of the auditory N1 and P2 components were estimated from N grand averages constructed using a leave-one-out approach. The time window for N1 amplitude analysis was defined based on these jackknife-derived peak latencies, together with visual inspection of the grand-average waveforms. Accordingly, N1 amplitude was averaged over a 116–136 ms time window after the auditory onset for both congruent and incongruent conditions. Additionally, P2 amplitude was measured in a 190–230 ms time window across different SOA conditions. Repeated-measures ANOVAs were performed on latency and mean amplitude measures, including two within-participant factors: Condition (congruent vs. incongruent) and SOA type (AV200, AV300, and AV600). To directly evaluate our a priori hypothesis, follow-up post hoc tests were conducted to further assess whether a congruency effect was observed within each SOA condition. Post hoc analyses of amplitudes and latencies were conducted using two-tailed tests, and *p*-values were corrected for multiple comparisons using the Holm–Bonferroni method. All statistical tests were conducted with a significance level of 0.05. In line with common practice in ERP studies, effects with p-values between 0.05 and 0.10 are described as trends toward significance for omnibus tests (i.e., main or interaction effects) (Keil et al., 2014). In addition, our previous study identified N1 congruency effects in bilateral, predominantly right-lateralized fronto-temporal regions among advanced English learners (Yan & Seki, 2024). To explore potential spatial differences in underlying neural processing within the N1 time window for IE learners, topographical analyses were conducted after confirming significant differences within the N1 time window, using a permutation test with 1000 randomizations. For the topographical analysis, we applied the false discovery rate (FDR) method (Benjamini & Hochberg, 1995) to account for multiple comparisons. Finally, as a supplementary analysis, we examined the relationship between neural responses and individual differences in English language proficiency by conducting two-tailed Pearson's correlations between participants' English language ability and ERP congruency effects (mean amplitudes) within the N1 time window.

3. Results

3.1. Results of behavioral experiment

The accuracy of the behavioral priming task was notably high, showing near-perfect levels of performance across the three SOA conditions (mean ± SD (AV200) = 98.18 ± 3.01%, mean ± SD (AV300) = 98.24 ± 2.04%, mean ± SD (AV600) = 98.41 ± 2.62%).

Regarding RTs, faster RTs were found in the congruent condition than the incongruent condition across the three SOAs. A condition × SOA repeated-measures ANOVA revealed a significant main effect of condition (F (1, 63) = 61.29, $p < 0.001$, $\eta^2 = 0.49$), but no significant effect of SOA condition (F (2, 63) = 0.11, $p = 0.90$, $\eta^2 = 0.00$) nor significant condition × SOA condition interaction (F (2, 63) = 1.63, $p = 0.20$, $\eta^2 = 0.05$). As shown in Fig. 2A, these results suggest a congruency effect on behavioral responses, indicating that the temporal relationships between letters and sounds did not influence the facilitation of congruent visual letters to the corresponding speech sound processing in L2.

3.2. Results of EEG experiment

3.2.1. Detection task

Participants demonstrated high accuracy in detecting targets during the EEG priming task. Mean detection rates [±SD] across the three SOA conditions were 97.02 ± 4.95% for AV200, 95.22 ± 4.53% for AV300, and 94.77 ± 3.98% for AV600. Moreover, the average of RTs [±SD] for target detection were 450.03 ± 62.25 ms for AV200, 454.08 ± 56.48 ms for AV300, and 460.90 ± 47.72 ms for AV600. To further assess the consistency of compliance among participants across all SOA conditions, we conducted two repeated-measures ANOVAs for detection accuracy and RTs throughout the three SOA conditions. The results revealed no significant main effects of SOA condition for either accuracy ($F(2, 57) = 1.40$, $p = 0.26$, $\eta^2 = 0.05$) or RT ($F(2, 57) = 0.19$, $p = 0.82$, $\eta^2 = 0.01$).

3.2.2. ERP data

Auditory ERP waveforms at electrode Cz, reflecting responses to auditory onset across the three SOA conditions, are illustrated in Fig. 3. Table 3 shows the grand-mean peak latencies and mean amplitudes of the N1/P2 at Cz electrode for both congruent and incongruent conditions across the three SOA conditions.

Differences in N1 and P2 peak latencies between congruent and incongruent conditions across the three AV types were examined using the condition × SOA type repeated-measures ANOVA. The analysis revealed no significant main effect of condition for either N1 ($F(1, 57) = 0.29$, $p = 0.59$, $\eta^2 = 0.01$) or P2 ($F(1, 57) = 0.51$, $p = 0.48$, $\eta^2 = 0.01$). Additionally, no significant main effect of SOA type was observed for either N1 ($F(2, 57) = 0.83$, $p = 0.44$, $\eta^2 = 0.03$) or P2 ($F(2, 57) = 0.38$, $p = 0.69$, $\eta^2 = 0.01$). Furthermore, no significant condition × SOA type interaction was found for either N1 ($F(2, 57) = 0.33$, $p = 0.72$, $\eta^2 = 0.01$) or P2 ($F(2, 57) = 0.34$, p = 0.72, $\eta^2 = 0.01$).

We conducted two repeated-measures ANOVAs to examine mean amplitudes for N1 and P2. For N1 amplitudes, we observed significant main effects of both condition ($F(1, 57) = 4.30$, $p = 0.043$, $\eta^2 = 0.07$), as well as a significant main effect of SOA type ($F(2, 57) = 25.40$, $p < 0.001$, $\eta^2 = 0.47$). Post hoc tests for main effect of SOA type revealed significant differences between AV200 and AV300 conditions, as well as AV200 and AV600 conditions. Detailed results for N1 amplitudes are provided in the Supplementary Materials (see Table S1). Although the condition × SOA type interaction did not reach significance, results indicated a trend toward significance ($F(2, 57) = 2.75$, $p = 0.072$, $\eta^2 = 0.09$). Post hoc pairwise comparisons examined the presence of a congruency effect, revealing significant differences in N1 amplitudes between congruent and incongruent conditions only in the AV300 condition ($t(57) = -2.91$, $p = 0.036$), with larger N1 amplitudes for congruent than incongruent trials. In contrast, no significant differences were found in the AV200 ($t(57) = -1.09$, $p = 1.00$) or AV600 condition ($t(57) = 0.40$, p = 1.00). To further quantify the evidence for congruency effects across SOA conditions in N1 amplitudes, Bayesian paired

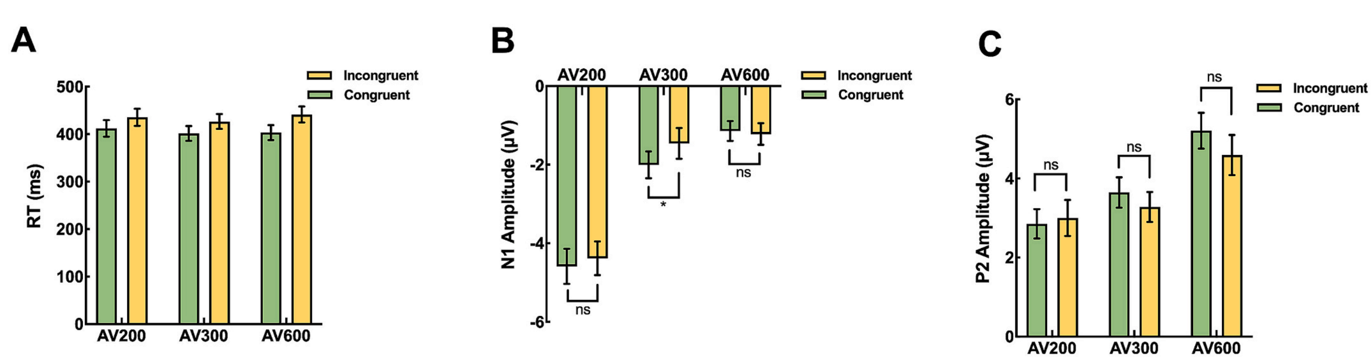

Fig. 2. Mean RTs in congruent and incongruent conditions across three SOA conditions in the behavioral experiment (**A**). Grand mean amplitude values of N1 (**B**) and P2 (**C**) components between congruent and incongruent conditions averaged over time across three SOA conditions in the EEG experiment. Error bars indicate standard errors.

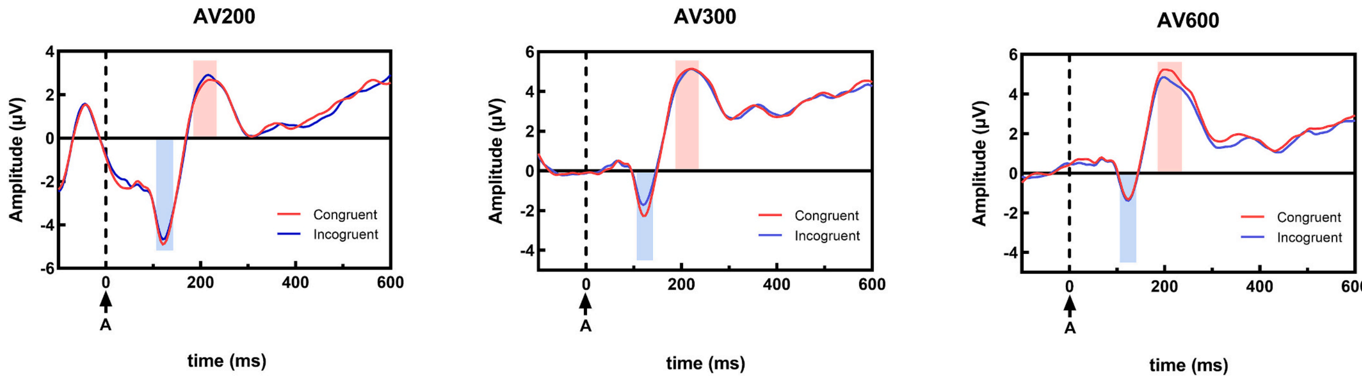

Fig. 3. ERP waveforms time-locked to auditory onset at electrode Cz, elicited by congruent (red) and incongruent (blue) conditions across three SOA conditions. "A" represents the onset of auditory stimulus. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Grand-mean (SEs) peak latencies and mean amplitudes of the N1/P2 at Cz electrode for two conditions (AVc, AVi) across three SOA conditions.

SOA condition	ERP component	Condition	Latency (ms)	Amplitude (μV)
AV200	N1	AVc	124.40 (1.97)	−4.59 (0.44)
		AVi	124.80 (2.18)	−4.38 (0.43)
AV300		AVc	123.80 (1.68)	−2.01 (0.34)
		AVi	121.80 (1.60)	−1.46 (0.39)
AV600		AVc	126.20 (2.44)	−1.14 (0.25)
		AVi	125.80 (2.31)	−1.22 (0.28)
AV200	P2	AVc	210.20 (4.78)	2.85(0.37)
		AVi	214.40 (5.56)	3.00 (0.45)
AV300		AVc	212.60 (4.32)	3.65 (0.38)
		AVi	212.20 (4.61)	3.28 (0.38)
AV600		AVc	207.00 (4.61)	5.21 (0.45)
		AVi	208.20 (5.36)	4.59 (0.51)

AVc = audiovisual congruent; AVi = audiovisual incongruent.

comparisons were conducted as a supplementary analysis. The results indicated strong evidence for a congruency effect in the AV300 condition ($BF_{10} = 29.70$), whereas evidence for the AV200 condition was weak ($BF_{10} = 0.63$), and the AV600 condition showed moderate evidence in favor of the null hypothesis ($BF_{10} = 0.18$).

For P2 amplitudes, we also found significant main effects of condition ($F(1, 57) = 4.72$, $p = 0.034$, $\eta^2 = 0.08$) and SOA type ($F(2, 57) = 6.12$, $p = 0.004$, $\eta^2 = 0.18$). Pairwise comparisons for main effect of SOA type indicated significant differences between the AV200 and AV600 conditions and between the AV300 and AV600 conditions. Additional results for P2 amplitudes are presented in the Supplementary Materials (Table S2). While the condition × SOA type interaction did not reach significance, a trend toward significance ($F(2, 57) = 3.11$, $p = 0.052$, $\eta^2 = 0.10$) was noted. Post hoc tests conducted to clarify the presence of congruency effects showed no significant differences in P2 amplitudes between congruent and incongruent trials for the AV200 ($t(57) = -0.68$, $p = 1.00$), AV300 ($t(57) = 1.66$, $p = 0.71$), or AV600 condition ($t(57) = 2.78$, $p = 0.081$).

Given post hoc and Bayesian analyses revealing the significant congruency effects during the N1 time window, a topographical analysis was performed to compare stimulus-evoked activity between congruent and incongruent trials. Fig. 4 presents the corresponding topographic representations of the auditory N1 component for the three levels of SOAs. Permutation tests revealed significantly greater negative activity for the AV300 condition ($p < 0.05$), specifically at the bilateral fronto-temporal sensors, for congruent trials compared with incongruent trials. In contrast, no significant differences were observed during the N1 time span for the AV200 or AV600 condition.

3.3. *Correlation between congruency effects and English proficiency*

As the significant N1 congruency effect was identified only in AV300, we conducted supplementary analyses to aid interpretation of this effect. Specifically, we examined whether the magnitude of the congruency effects in auditory N1 mean amplitudes was associated with English proficiency levels. Given the strong positive correlations among the English proficiency subtest scores, it is challenging to select one specific test to represent overall English proficiency. To address this, we performed a principal component analysis on all English proficiency subtest scores to identify the principal components explaining the highest proportion of variance. Table 4 lists the principal component loadings for each variable. The analysis revealed that the first principal component, with a contribution of 40.78%, likely indicated comprehensive English proficiency. Accordingly, the first component was used as a composite index of overall L2 proficiency in subsequent analyses. Notably, in contrast to our previous study, the phonological awareness measures showed a substantially smaller loading on the first principal component, possibly suggesting a different contribution of subskills to overall proficiency in the present sample of intermediate learners. Separately, based on the results of the topographical analysis, we chose to focus on the C3, Cz, and C4 electrodes. No significant correlations were found between L2 proficiency levels and the N1 congruency effect in either the left or right hemisphere (C3: $r = 0.20$, $p = 0.40$; Cz: $r = 0.17$, $p = 0.49$; C4: $r = 0.35$, $p = 0.13$). In addition, our analysis did not yield any significant correlation between ASRS scores and ERP congruency effects, indicating that ADHD-like traits may not modulate the congruency effect on N1 responses (C3: $r = -0.29$, $p = 0.22$; Cz: $r = 0.17$, $p = 0.48$; C4: $r = -0.31$, $p = 0.18$). Finally, we additionally examined the relationship between L2 proficiency and the N1 congruency effects at electrode Cz in the AV200 and AV600 conditions. These analyses likewise revealed no significant correlations (AV200: $r = 0.23$, $p = 0.33$; AV600: $r = -0.25$, $p = 0.29$).

4. Discussion

Our study aimed to investigate how variations in the temporal

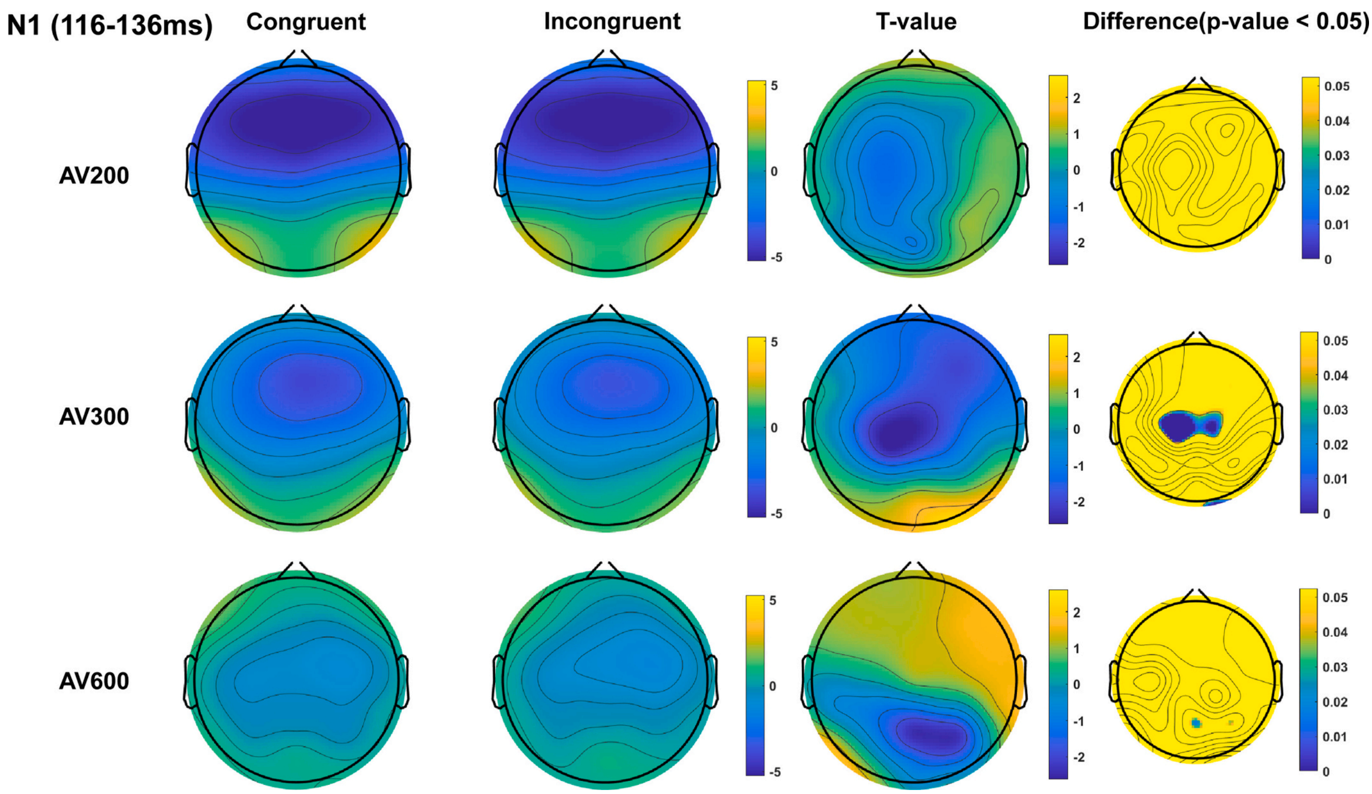

Fig. 4. Topographical analysis illustrates averaged ERPs across electrodes in the N1 time window for the congruent and incongruent conditions across three SOA conditions. The right column shows permutation-based *p*-value distributions for statistical comparisons within conditions.

Table 4
Summary of principal component analysis: principal component loadings of each variable contributing to two identified components.

Variables	Principal Component (PC) 1	2
TOWRE II Sight Word Efficiency	**0.79**	−0.09
TOWRE II Phonemic Decoding Efficiency	**0.91**	0.09
WJ III LWI	**0.61**	0.38
WJ III WA	**0.73**	0.19
WJ III RF	**0.81**	−0.01
WJ III SS	0.34	**0.56**
CTOPP-2 Blending Words	−0.04	**0.93**
CTOPP-2 Blending Nonwords	−0.12	**0.80**
CTOPP-2 Segmenting Nonwords	0.24	**0.86**

Boldface values indicate the testing measurement contributing to the principal component above 0.5, which is deemed to have a stronger correlation with the component. TOWRE = Test of Word Reading Efficiency; WJ = Woodcock–Johnson; LWI = Letter–Word Identification; WA = Word Attack; RF = Reading Fluency; SS = Spelling of Sounds; CTOPP-2 = Comprehensive Test of Phonological Processing–Second Edition.

relationship between cross-modal input influence L-SS integration, as indicated by the presence of congruency effects among native Japanese speakers with intermediate level of English proficiency. To achieve this, we measured both behavioral performance and ERP responses using an AV priming paradigm.

4.1. Behavioral measure

The results of the behavioral measure revealed significant differences in RTs between congruent and incongruent conditions across all three SOA conditions, showing an AV behavioral benefit with shorter RTs observed in the congruent condition. These results are consistent with those found in previous research (Clayton, West, Sears, Hulme, & Lervåg, 2020) as well as those from our earlier study (Yan & Seki, 2024) with the use of a priming paradigm. Consequently, we can assume that the behavioral congruency effect persists regardless of temporal asynchrony between modalities. However, behavioral results alone provide limited insights into the detailed mechanisms underlying L-SS integration process. Therefore, further analysis of the details in the ERP experiment is required.

4.2. ERP measure

4.2.1. Congruency effects in N1 amplitudes within each SOA condition in intermediate English learners

First, we observed a significant main effect of condition (congruent vs. incongruent) within the auditory N1 time window, indicating reliable amplitude differences between these two conditions. However, this main effect alone does not specify the direction of the difference and therefore does not directly establish the presence of a congruency effect. Meanwhile, a significant main effect of SOA was observed as well. As SOA was manipulated between blocks and baseline correction was time-locked to auditory onset, differences in the timing of visual cues across SOA conditions may have introduced baseline-related variability. This variability may have contributed to amplitude differences across SOA. Importantly, such baseline-related variability may also have reduced sensitivity to detect interaction effects. In line with this possibility, the Condition × SOA interaction did not reach statistical significance, suggesting that the magnitude of the difference between congruent and incongruent conditions did not reliably vary across SOA types.

To determine whether a congruency effect was observed within each SOA conditions, we therefore conducted post hoc and complementary Bayesian analyses. These analyses revealed an N1 congruency effect only in the AV300 condition for IE learners. In the AV200 condition, our results revealed no significant congruency effects either in the N1 mean amplitude or in the scalp distribution, partially replicating findings from our previous work (Yan & Seki, 2024). In contrast, the AV300 condition showed significant congruency effects in both N1 amplitude and topographical distribution, showing a larger N1 amplitude evoked by the congruent condition, specifically at bilateral frontotemporal electrodes. The observed region, consistent with previous fMRI findings (e.g., Blau et al., 2008; Blau et al., 2009; van Atteveldt et al., 2004), may correspond to the auditory cortex, a region involved in L-SS integration. Thus, the increased N1 amplitude in the congruent condition may reflect modulation of auditory processing by visually presented letters when they are phonologically congruent with the speech sounds. Meanwhile, according to the feedback hypothesis proposed by van Atteveldt et al. (2009)—a theoretical model describing the neural mechanisms underlying L–SS integration—early modulation in the auditory cortex may result from successful integration between visual letters and auditory sounds in the STC, where congruent visual input activates L-SS associations that, in turn, influence auditory cortical activity. Notably, this congruency effect occurred within the N1 time window, reflecting the early stages of auditory processing. The early timing of this effect indicates that L-SS integration may occur rapidly and automatically. Therefore, the N1 congruency effect observed in the AV300 condition among IE learners may be interpreted as neural evidence for early and automatic L-SS integration. Nevertheless, to better understand the impact of individual difference, we found that unlike findings from advanced learners in our previous study, the congruency effects in the AV300 condition did not correlate with English proficiency scores. This may be due to the relatively limited variability in L2 proficiency or heterogeneity in linguistic subskills among intermediate learners, making it difficult to detect proficiency-related modulation of congruency effects. Further investigation will be needed to clarify this issue.

By contrast, no congruency effect was observed in the AV600 condition. This finding may be explained by notion that when the SOA becomes too long, the integration of AV information may be less likely to occur. A relevant study by Leone and McCourt (2015) examined temporal dynamics in AV perception using simple light and tone stimuli. Their study suggests that excessively long SOAs may impair AV integration, potentially due to the perception of the auditory and visual stimuli as separate events. Although the task and stimulus types in that study differ considerably from the present paradigm—ours involving complex linguistic stimuli—the underlying temporal principle appears to be generalizable. In the present study, the absence of a congruency effect in the AV600 condition may support this interpretation. While this result should be interpreted cautiously, it suggests that AV integration processes, even for linguistic stimuli, may be constrained by the temporal integration window.

Taken together, our results show that a significant N1 congruency effect was observed in the AV300 condition in native Japanese speakers with intermediate English proficiency, whereas no reliable congruency effects were found in the AV200 or AV600 conditions. Overall, our findings highlight the importance of temporal alignment between letters and speech sounds in L-SS integration for IE L2 learners.

4.2.2. Proficiency-dependent temporal dynamics in L-SS integration

In our previous study (Yan & Seki, 2024), advanced learners exhibited N1 congruency effects in the AV200 condition, whereas no such effect was observed in IE learners. In contrast, the current study found that learners with IE proficiency level showed larger N1 amplitudes in the congruent condition only for the AV300 condition. To facilitate comparison with our previous findings, we descriptively examined the magnitude of the N1 congruency effect taking cross-study differences into account. The effect observed in the AV300 condition for IE learners was of moderate size (Cohen's $d_z \approx 0.62$), comparable to that reported for advanced learners in the AV200 condition (Cohen's $d_z \approx 0.49$).

When considered together, these findings may suggest that the temporal characteristics of L-SS integration may differ across learners with different L2 language proficiency. Specifically, these results may provide evidence that IE learners potentially require slightly longer intervals to successfully integrate English letters and sounds. Previous research has shown that multisensory experience—such as bilingualism (Bidelman & Heath, 2019) or musical training (see a review, Noppeney & Lee, 2018)—can narrow the width of TBW, reflecting greater temporal precision in multisensory integration. Consistent with this view, Du et al. (2023) reported that native Chinese speakers with intermediate English proficiency exhibited enhanced bimodal MMN responses in the AV200 condition following short-term English letter-sound training, but not symbol-sound training, whereas no such enhancement was observed prior to training. These findings suggest that targeted L2 L-SS training experience may narrow the width of TBW, thereby facilitating early integration. Importantly, the current paradigm, as in previous studies manipulating SOAs in L-SS integration (e.g., Froyen et al., 2008), employed visual-leading AV stimuli. Given that visual letters precede auditory sounds, the efficiency of bottom-up activation of L-SS associations may play a critical role in determining the temporal window for successful integration. Therefore, another plausible explanation is that the slower or less automatic activation of L-SS association may therefore delay the emergence of congruency effects. In line with this, previous studies have indicated that congruency effects of ERP amplitudes between congruent and baseline conditions have been observed in children with dyslexia in the AV300 condition (Du et al., 2022), and in a broader time window (i.e., 1000 ms) between the presentation of letters and sounds (Nash et al., 2017). These findings suggest that the activation of L-SS associations may require more time in less proficient readers, such as those with dyslexia. Accordingly, in the present study, longer temporal requirements in intermediate L2 learners are interpreted as a tentative possibility, potentially reflecting slower or less automatic activation of L-SS associations.

In summary, our findings may provide evidence on how L2 proficiency relates the temporal coordination between letters and speech sounds during L-SS integration. Notably, IE learners appeared to exhibit congruency effects at a slightly longer temporal interval than advanced learners, suggesting that lower-proficiency readers may require more time to achieve successful integration. In such contexts, the present results potentially suggest that the efficiency or timing of L-SS integration may vary as a function of language experience (i.e., L2 proficiency levels), although further investigation is required to substantiate this interpretation. More broadly, further research is warranted to clarify whether atypical or extended temporal integration are associated with reading difficulties in L2 contexts.

4.2.3. Topographical differences and proficiency-related lateralization

In the present study, IE learners exhibited N1 congruency effects with a more bilateral distribution over the frontotemporal regions. In contrast, findings from our previous study (Yan & Seki, 2024) in the advanced group learners indicated a more right-lateralized N1 congruency effect. This topographical difference suggests that L2 proficiency may influence the hemispheric distribution of neural responses during L-SS integration. Previous neuroimaging studies on L2 learning have consistently shown changes in lateralization as L2 proficiency increases (Sander et al., 2023; Wei et al., 2024; Xiang et al., 2015). Specifically, early-stage L2 learners tend to show a more bilateral distribution, while advanced learners typically demonstrate greater right hemisphere involvement. These findings align with our topographical findings and support the notion that lateralization changes with increasing L2 proficiency, reflecting the dynamic plasticity of the brain in adapting to language learning demands.

4.3. Limitations

A limitation of this study is the lack of investigation into the impact of SOA on advanced English learners. This limitation has two important aspects. First, the effect of the temporal relationship between visual and auditory inputs on L-SS integration needs investigation among advanced learners. In our previous study (Yan & Seki, 2024), advanced learners exhibited congruency effects only in the AV200 condition. However, data from advanced learners for other SOA conditions (e.g., AV300 and AV600) are currently lacking, making it challenging to estimate the impact of TBW width on L-SS integration in this group. By examining congruency effects using various SOAs in advanced learners, future research could investigate whether the size of the TBW changes with increasing reading proficiency. Second, it is necessary to include both intermediate and advanced learners to better understand the correlation between congruency effects in the AV300 condition and English proficiency. In the present study, no significant correlation was found between congruency effects and English proficiency. Additionally, in the current study, the contribution of all English proficiency subtest scores to the first principal component is markedly differed from that observed in our previous study. These discrepancies may be attributable to the relatively restricted range of proficiency within the intermediate learner group, which reduces variability and may hinder detection of proficiency-related modulation of congruency effects. Moreover, the influence of a relatively small sample size cannot be ruled out. Therefore, future studies should include both intermediate and advanced learners with larger samples to systematically examine congruency effects across the AV200, AV300, and AV600 conditions. Additionally, future work may benefit from including non-verbal intelligence as a covariate to further disentangle language-specific and domain-general contributions to L-SS integration.

Given that preceding visual inputs (e.g., mouth movements) may facilitate auditory processing at both behavioral and neural levels (e.g., Navarra, Hartcher-O'Brien, Piazza, & Spence, 2009; van Wassenhove, Grant, & Poeppel, 2005), and are more closely aligned with typical reading processes, the present study focused on visual-leading SOAs. An important open question, however, is whether similar patterns would emerge under auditory-leading conditions. Future studies may examine whether reversing stimulus order alters integration dynamics across different levels of language proficiency.

5. Conclusions

To conclude, the present study aimed to investigate how different SOAs relate to L-SS integration during L2 reading among native Japanese speakers with IE proficiency levels. A behavioral priming experiment revealed shorter RTs in response to the congruent condition across all three SOA conditions, indicating that a general facilitation effect of congruency occurred regardless of the degree of temporal coincidence between letters and speech sounds. In the EEG experiment, we found the congruency effect on the early auditory N1 response in the bilateral frontotemporal scalp region during L2 reading, most prominently in the AV300 condition. Based on our findings, we speculate that the temporal relationship between auditory and visual inputs modulates the N1 congruency effects in native Japanese speakers with IE proficiency. In particular, a slightly longer SOA in a visual-first temporal task may facilitate the successful integration of English letters and speech sounds, under conditions in which automatic activation of L-SS associations is less efficient. Altogether, the results highlight the importance of temporal dynamics of AV inputs in L2 L-SS integration and suggest that the efficiency or timing of L-SS integration may be modulated by the level of L2 proficiency. These findings deepen our understanding of the neural mechanisms underlying L2 reading and suggest that proficiency levels related to temporal integration may contribute to variability in L2 reading performance. More broadly, the results may provide a theoretical basis for future research examining the role of atypical L-SS integration in L2 reading difficulties, particularly among learners with limited proficiency.

CRediT authorship contribution statement

Dongyang Yan: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Ayumi Seki:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Written informed consent was obtained from all adult participants in the current study.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bandl.2026.105816.

Data availability

The datasets analyzed in the current study are available from the corresponding author upon request.

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